

Name of the Student _____

Scholar Number _____

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION-NOVEMBER 2023

Course: B.Tech

Semester: III

Subject Name: Measurement and Instrumentation

Subject Code: EE-214

Time Duration: 3 Hrs

Maximum Marks: 60

Subject Coordinator Names: Dr. Sushma Gupta & Dr. More Raju

S.No	Question	Marks	CO
1	A) A 1000/5 A, 50 Hz current transformer has a secondary burden comprising a non-inductive impedance of 1.6Ω . The primary winding has single turn. Calculate the flux in the core and ratio error at full load. Neglect leakage reactance and assume the iron loss in the core to be 2 W at full load. The magnetizing MMF is 100 A.	5	CO-5
	B) Define the following terms as used for instrument transformers. i. Transformation ratio ii. Nominal ratio iii. Turns ratio iv. Burden v. Ratio correction factor	3	CO-5
	C) The value of a resistor is specified as $300 \Omega \pm 10\%$ by a manufacturer. Find the limits of the resistance which the value is guaranteed.	2	CO-1
2	A) Explain the construction and principle of operation of single phase energy meters.	5	CO-2
	B) A moving coil instrument whose resistance is 25Ω gives a full scale deflection with a current of 1 mA. This instrument is to be used with a manganin shunt to extend its range to 200 mA. Calculate the error caused by 20°C rise in temperature when i. Copper moving coil is connected directly across the manganin shunt ii. A 80Ω manganin resistance is used in series with the instrument moving coil. The temperature co-efficient of copper is $0.004/^\circ\text{C}$ and that of manganin $0.00015/^\circ\text{C}$.	5	CO-2
3	A) Explain how power can be measured in a three phase circuit with the help of two wattmeters. Illustrate your answer with help of a phasor diagram for a balanced star connected load.	5	CO-2
	B) Define Blondel's theorem.	2	CO-2
	C) Two wattmeters are connected to measure the input to a balanced 3 phase circuit indicate 3000 W and 500 W respectively. Find the power factor of the circuit	3	CO-2

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	i. when both the readings are positive. ii. when the latter reading is obtained after reversing the connections to the current coil of first instrument.		
4	A) Explain Schering Bridge with circuit diagram and phasor diagram. Derive the expression for unknown capacitance, loss in capacitance and Dissipation factor. Enumerate advantages and disadvantages also. B) A Kelvin double bridge each of the ratio arms $P = Q = p = q = 1000 \Omega$. The emf of the battery is 100 V and a resistance of 5Ω is included in the battery circuit. The galvanometer has a resistance of 500Ω and the resistance of the link connecting the unknown resistance to the standard resistance may be neglected. The bridge is balanced when the standard resistance $S = 0.001 \Omega$. i. Determine the value of unknown resistance. ii. Determine the current through the unknown resistance at balance. iii. Determine the deflection of the galvanometer when the unknown resistance R is changed by 0.1 % from its value at balance. The galvanometer has a sensitivity of $200 \text{ mm}/\mu\text{A}$.	5	CO-4
5	A) Explain how thermistor is more sensitive temperature transducer compared to RTD. Draw and explain voltage-current and current-time characteristics of thermistor. B) If you have to measure depth of Ocean and level of petroleum in tank then which techniques should be adopted? Explain each one in detail.	5	CO-4
		5	CO-3
6	A) Describe the Bourden Tube, Bellows and Diaphragm pressure transducer. B) Explain strain gauge theory. Derive the expression of gauge factor.	6	CO-3
		4	CO-3